CONCEPT RESEARCH PROPOSAL

From motor brain-computer interfaces to a closed-loop cognitive copilot

A phased program to test whether a human and an external AI can share a narrow, auditable channel for higher-level work — without claiming a merged mind.

Document type. Pre-clinical and early clinical research concept. Not an IND, IDE protocol, or device submission.
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Status. The end-state described in popular form — a Neuralink that melds a person with a model such as Grok — does not exist and is not the near-term endpoint of this proposal.

Prepared as a planning document. Public facts about Neuralink's 2026 trial are used only as a baseline. They are company-reported and not independently re-analyzed here.

1. Purpose

The question is whether a person can do harder intellectual work by coupling their brain, through a brain-computer interface (BCI), to an external model, so that retrieval, draft structure, and consistency checks arrive with less friction than typing. The popular version of that question is a shared brain: one process, human neurons and model weights fused. This proposal rejects that as the research target. It replaces it with a narrower, testable system: a closed-loop cognitive copilot. The person remains the agent. The model remains a tool. The implant, if it is ever used in humans, is only a low-latency channel under explicit consent and continuous override.

Success is not a higher IQ and not a subjective feeling of merger. Success is a measured gain on pre-registered tasks — multi-step reasoning, source-checked writing, error detection — relative to the same person using a keyboard and the same model, after subtracting time spent correcting the channel.

2. What exists now

As of October 2026, Neuralink's N1 is an investigational, fully implantable wireless BCI evaluated in the PRIME study and related work. It records through 1,024 electrodes on 64 threads. The published use is decoding movement intention: cursor control, device use, games, browsing. On 1 October 2026 the company reported more than 50,000 hours of neural recordings from trial participants and an individual cursor-control result of 11.32 bits per second, with a participant median around 10 bits per second. Calibration burden has fallen in company reports. Enrollment figures cited publicly earlier in 2026 were on the order of 21 participants worldwide. The device is not commercially approved. Speech restoration remains investigational. Nothing in that record is a read of concepts, a write of ideas into cortex, or a link to a language model.

Other clinical BCIs (Utah array, ECoG, stentrode, and speech neuroprostheses) have restored communication or movement intent. Speech BCIs have reached useful words-per-minute in single participants. None has demonstrated transfer of abstract reasoning into or out of a model as a shared cognitive state. Ten to eleven bits per second is enough to beat some cursor tasks. It is not enough to stream thought.

3. The gap, stated as engineering constraints

A meld fails on four independent constraints. A research program has to retire them one at a time, or stop.

Constraint	Current reality	What would have to be true
Read bandwidth	Motor decode at about 10 bits/s in the best reported Neuralink cursor result.	Stable decode of intended speech or discrete cognitive commands at a rate that beats typing for the target task.
Write path	No safe, specific way to insert a proposition into human cortex.	A write channel limited to sensory or prosthetic feedback the user can ignore. Not memory implantation.
Identity of content	Models confabulate. Decoders drift.	Every model output is labeled as external. The user can reject it in one action. Drift triggers automatic fallback to manual input.
Agency	A fused process has no clear author of a decision.	The human is the only agent who can commit an action. The model cannot initiate external effects.

The proposal therefore does not aim at "sharing brain and algorithm abilities" as a single mind. It aims at a copilot channel whose errors are visible.

4. Specific aims

Aim 1 — Define the task, not the fantasy.

Pre-register three task families a copilot could plausibly help within ten years: (1) drafting and revising a technical memo against a fixed source packet; (2) detecting contradictions in a long argument; (3) holding a multi-step plan while the person is interrupted. Primary endpoint is accuracy per minute of engaged time versus keyboard-plus-same-model. Secondary endpoints are error-correction time, subjective workload, and rate of accepted false statements.

Aim 2 — Exhaust the non-invasive ceiling.

Before any new implant, measure the same tasks with speech, gaze, EMG, and high-density EEG as the input channel to the same model. If a non-invasive channel captures most of the gain, an implant is not justified for this use. This aim is a kill criterion, not a formality.

Aim 3 — Speech-intent decode as the only near-term neural read.

In participants who already have a clinical motor or speech BCI for a medical indication, test whether intended speech or a small command vocabulary can drive the copilot faster than their existing assistive method, without degrading the clinical function the implant was approved to study. No healthy-volunteer implants for enhancement.

Aim 4 — External model with a hard agency boundary.

Couple the decoder only to a tool-using model that retrieves, drafts, and critiques. The model may not send messages, move money, or change a record unless the human confirms on a separate, non-neural control. Log every suggestion and every acceptance.

Aim 5 — Sensory write only, and only if Aims 2–4 show a deficit that feedback would fix.

Do not attempt to write semantic content into cortex. If a write experiment is ever justified, limit it to a discriminable cue (a brief cortical or sensory percept meaning "suggestion ready" or "low confidence"). Stop if the cue is confused with the user's own thought.

5. Phased plan

Phase	Years	Work	Stop rule
0. Framing	0–1	Ethics board, task design, threat model, public pre-registration. No surgery.	Aims not measurable, or consent model cannot separate clinical need from enhancement.
1. Non-invasive copilot	1–3	Same model, speech/gaze/EEG input, n large enough for a within-person crossover.	Gain over keyboard is under a pre-set margin once correction time is counted.
2. Clinical add-on	3–7	Add-on protocol in existing BCI recipients with a medical indication. Speech-intent or motor commands only.	Any increase in adverse events versus the parent trial, or clinical function worsens.
3. Narrow feedback	7–12	Cue-level feedback only, in the same clinical population, if Phase 2 shows a specific bottleneck.	Users cannot reliably tell cue from endogenous sensation, or agency ratings fall.
4. Reassessment	12+	Only if prior phases beat the non-invasive ceiling on the primary endpoint.	Default is not to build a higher-bandwidth cognitive implant.

Phase 4 is a decision gate, not a promise. A merged cortex-and-weights system is outside this plan. It would require a write bandwidth and a theory of cortical meaning that do not exist.

6. Technical work packages

6.1 Decoder

Treat the neural signal as a noisy intended-speech or intended-command channel. Use self-supervised pretraining on longitudinal recordings to reduce daily calibration, following the direction Neuralink reported in 2026 for motor decode, but evaluate on held-out days and held-out tasks. Report bits per second, words per minute, and word error rate, not anecdotes. Publish drift curves. A decoder that needs ritual recalibration will lose to a keyboard.

6.2 Model boundary

The model is interchangeable. Grok is a reasonable test system because it already does retrieval and drafting; the experiment must also run on at least one other model so results are not vendor-specific. The interface sends text or a structured command, not raw spikes, across the trust boundary. Raw neural data stay on the medical side of the system and are not used to train a public model. Suggestions return as text plus a confidence and a source list.

6.3 Override and logging

One neural or switch gesture aborts the suggestion and returns control to the clinical decoder. Acceptance of a suggestion is a separate confirm act. Logs are readable by the participant. There is no hidden personalization that the participant cannot export or delete.

6.4 What is out of scope

- Memory implantation, dream writing, or any claim to insert a belief.
- Continuous subconscious coupling. The channel is on only when the participant opens it.
- Healthy enhancement implants. Indication, if any human neural work occurs, is an existing medical BCI need.
- Autonomy for the model: no unconfirmed external actions.
- IQ as an endpoint. The endpoint is task throughput and error rate.

7. Ethics, safety, and governance

The enhancement framing is the main ethical hazard. A device justified as restoring communication and then quietly retargeted as cognitive merger would fail consent. This proposal keeps those uses separate. Participants in any neural phase must already be in a clinical BCI study for a therapeutic indication. They can leave the copilot protocol without losing the clinical device function. Incidental findings in neural data are handled under a pre-written policy.

Risks to state in the consent: surgical risk of the parent trial (hemorrhage, infection, seizure, device failure); decoder error accepted as one's own intention; model hallucination accepted under time pressure; privacy of neural recordings; coercion if a workplace treats the channel as mandatory. Agency ratings and a "was this my thought?" probe are safety endpoints, not marketing metrics. A fall in the participant's ability to detect model errors stops the study.

Governance: institutional review board before Phase 1; FDA IDE or equivalent only if Phase 2 alters an investigational device; an independent data monitor; pre-registration; no exclusive claim on a participant's neural data by the model vendor. Dual-use review covers any decode that could be repurposed for covert interrogation. That use is forbidden in the protocol.

8. Evaluation

Primary comparison is within person: copilot channel versus keyboard, same model, counterbalanced. A result counts only if accuracy does not fall and throughput rises after correction time. Pre-set the smallest effect worth an implant: if Phase 1 captures it, Phase 2 does not open. Report failures. Company-style single-participant records are allowed as feasibility, not as evidence of a general cognitive gain.

9. Team and rough resources

Minimum team for Phases 0–1: a BCI scientist, a speech neuroscientist or motor-control specialist, a human-computer interaction researcher, a statistician, an ethicist with device experience, and clinical advisors who are not employed by the implant vendor. Phase 2 cannot start without a neurosurgical site already running an approved BCI trial and a regulator in the loop. Budget is dominated by trial operations, not by model API cost. A university-plus-clinical site structure is a better fit than a single company lab, because the kill criteria have to be enforceable against the group that wants the implant to succeed.

10. Recommendation

Bring the idea into reality by shrinking it. Build and publish the non-invasive copilot comparison first. Treat today's motor BCI — on the order of 10 bits per second of cursor control in the best 2026 Neuralink reports — as a clinical communication tool, not as a prototype mind. Add a model only as a labeled, abortable suggestion channel. Do not fund a write-into-memory program under this rationale. If the non-invasive channel already makes the person more effective on the pre-registered tasks, that is the result, and it does not require a new brain implant.

Sources for the 2026 baseline

Neuralink, 1 October 2026 blog report of more than 50,000 hours of participant neural data and a 11.32 bits-per-second individual cursor result (company-reported; contemporaneous trade coverage including TeslaNorth and Interesting Engineering). PRIME study public descriptions of the N1 implant (1,024 electrodes, 64 threads) and of movement-intention decode. Enrollment snapshots reported earlier in 2026 are not a current census. These figures are a floor for honesty in the proposal, not an endorsement of any device.

This document is a research concept. It is not medical advice, not an offer to build a device, and not a claim that a human–AI merged mind is feasible.